

# Arjun Prasad

Senior Backend Engineer | Distributed Systems & Payments Infrastructure

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## SUMMARY

Backend engineer with 4.5+ years building high-throughput payment systems at PhonePe -- India's largest digital payments platform (500M+ users). Currently on the UPI Core Transactional team handling 20,000+ RPS and 500M+ daily transactions. Previously owned BBPS infrastructure processing 4M+ transactions/day. Built greenfield microservices, created a platform-wide async framework cutting infra costs 45%, and led Active-Active DC migration with circular replication. Strong focus on reliability, scalability, and end-to-end ownership.

## TECHNICAL SKILLS

|                        |                                                                    |
|------------------------|--------------------------------------------------------------------|
| <b>Languages:</b>      | Java (8-17), C++, SQL                                              |
| <b>Frameworks:</b>     | Dropwizard, Spring Boot, Hibernate, JUnit, Maven                   |
| <b>Data Stores:</b>    | MariaDB (Sharded), Aerospike, MySQL                                |
| <b>Infrastructure:</b> | Microservices, Active-Active DC, Docker, CI/CD, Nginx, Linux       |
| <b>Messaging:</b>      | RabbitMQ, MemQ, Event-driven Architecture                          |
| <b>Practices:</b>      | System Design, Distributed Systems, DB Sharding, REST APIs, DR/BCP |

## EXPERIENCE

### PhonePe -- UPI Core Transactional Team

Mar 2026 - Present

*SDE-2 (Backend) | 20,000+ RPS | 500M+ UPI transactions/day*

- Operating on India's highest-traffic payment rail -- 20K+ RPS peak, mission-critical availability for 500M+ daily UPI transactions
- Building E2C refund status tracking for 2-2.5 lakh daily refunds -- users currently have zero in-app visibility (must check bank statements)
- Owning NPCI contract architecture, UDIR (UPI Dispute Resolution) flows, and external PSP integrations (GPay to PhonePe)

### PhonePe -- BBPS Pod

Feb 2022 - Feb 2026

*Software Engineer to SDE-2 (Backend) | 500 RPS peak | 4M+ billpay transactions/day*

#### UPMS -- Greenfield Microservice (0 to 1)

- Independently designed and shipped a new microservice end-to-end -- sharded DB schema, proxy layer, recon system, biller kill-switches -- zero to 500+ RPS in 5 months
- Replaced polling with push-based presentment driving 20% transaction growth; completed NPCI certification and monolith data migration

#### Axon -- Async Messaging Framework (45% Infra Cost Reduction)

- Created a reusable framework abstracting RMQ/MemQ with unified publish API, MemQ-to-RMQ fallback, and dynamic worker buckets -- adopted org-wide
- Reduced per-service queue integration from days to minutes; 45% infrastructure cost reduction across PhonePe

#### Active-Active Data Center Migration

- Led migration: redesigned DB schema for circular replication, migrated async workers to MemQ (DC-aware), validated cross-DC routing
- Achieved zero-downtime DC failover; blast radius reduced from 100% to ~50% (single DC scoped)

#### Status Check Pipeline Overhaul

- Reduced stuck transactions from 20,000+/day to <100/day; resolution time from 9+ hours to <10 min; eliminated wrong refund risk

#### Other Contributions

- Piped Gas Meter Reading: contract to production in 1 week, demonstrated at Global Fintech Festival 2025, PhonePe-exclusive
- Paylink migration (+15% transactions, zero downtime) | Loan API 1.1 (led NPCI spec compliance independently)
- Launched 6 bill categories; created SOP eliminating eng dependency | Mentoring 3 engineers | 117 code reviews | 6 DR drills

### Samsung Research Institute -- Software Engineer

Jul 2021 - Feb 2022

- Reduced audio device setup time by 30% through Bluetooth module optimization (1,000+ devices)
- Samsung SWC Professional certification (Data Structures & Algorithms)

## EDUCATION

### B.Tech in Computer Science & Engineering

2017 - 2021

National Institute of Technology, Delhi

## RECOGNITION & IMPACT

- Received "Distinguished" rating (highest tier) at PhonePe -- "Completed the implementation flawlessly and quickly" -- Engineering Manager
- Scale progression: 500 RPS (BBPS, built from scratch) to 20,000+ RPS (UPI Core) -- consistently entrusted with increasing scope
- Built 2 greenfield systems (UPMS service + Axon framework) and led 1 major infra migration (Active-Active DC) -- all in production at scale
- 345+ merge requests | 605 tasks delivered | 356 architecture documents authored | 4 consecutive strong performance cycles